

EVA N. BURGOS

Greater Houston, TX Area, 77345 □ 281-906-4990 □ natalie.e.burgos@gmail.com

EDUCATION

Bachelor of Science, Biomedical Engineering – THE UNIVERSITY OF TEXAS AT DALLAS – Richardson, TX
CSWA (Certified SolidWorks Associate), CSWP (Certified SolidWorks Professional), CSWPA-DT (Certified SolidWorks Professional Advanced Drawing Tools), CSWPA-SM (Certified SolidWorks Professional Advanced Sheet Metal), CSWPA-WD (Certified SolidWorks Professional Advanced Welding).

RESEARCH PROJECTS & EXPERIENCE

Internship: Clinical Engineering, R&D Department, Argon Medical Devices, Plano, TX 1/2025-5/2025

- **Synthetic Clot Development:** Engineered a synthetic clot model to simulate real-world vascular conditions for testing thrombectomy devices, refining its material composition to achieve physiologically relevant behavior.
- **TIPS Liver Block Fabrication:** Created Transjugular Intrahepatic Portosystemic Shunt (TIPS) liver blocks using custom silicone formulations to enhance procedural training and product evaluation in interventional radiology applications.
- **Thrombectomy Device Prototyping & Verification Testing:** Supported iterative prototyping of a thrombectomy catheter system; performed bench-level performance testing (trackability, aspiration efficiency) to guide design refinement and regulatory documentation.

Co-Op: New Product Development Engineering, R&D Department, STERIS, Conroe, TX 8/2024-12/2024

- **Sample Container Holder Project:** Designed and fabricated a custom holder for securing sample containers during laboratory testing. Conducted background research to identify optimal materials, creating a CAD design using SolidWorks and 3D printing and prototypes. The project involved iterative testing and adjustments to enhance durability and fit, resulting in a cost-effective and reusable holder that improved workflow efficiency and was widely adopted by the R&D team.
- **Scope Buddy Project:** Leads the design and development of a new O-ring to improve the Scope Buddy 1700 device's seal and water flow. Takes charge of prototyping, performance testing, and design refining, coordinating efforts with the engineering team and presenting critical updates to senior engineers.
- **Syringe Recall and Validation Project:** Collaborates with the marketing team to assess and recommend suitable syringe replacements. Supports the testing and validation of syringes following an FDA recall, which includes updating part numbers and IFUs to meet compliance.
- **Clamshell project:** Designed and prototyped a "clamshell" for an LED project using SolidWorks to ensure consistent curing of valves, addressing misalignment issues in production. Collaborated with senior engineers measuring various valve types, considered wavelength differences, and iterated designs while maintaining compatibility with existing systems.

Project: "ASSERT: A Platform Technology for Rapid Electromechanical Sensing of Soil Ammonium." **Undergraduate Researcher, University of Texas at Dallas, 8/2022-5/2024.**

- Co-authored the following publication based on the project: "ASSERT: A platform technology for rapid electromechanical sensing of soil ammonium," American Chemical Society, University of Texas at Dallas, 2024.
- Collaborated with a Ph.D. student on a year-long project that focused on measuring the stability of soil in order to correct an imbalance in the soil ecosystem affecting carbon sequestration, plant-available nutrients, and microorganisms (through the use of sustainable agriculture that uses the soil efficiently with minimal fertilizer and crop rotation).
- Completed a detailed literature review, conducted the preparation and drop casting of the ion-selective electrode onto the working electrode, and calibrated the sensor response against 3 different soil textures (Clay, Sandy Loam, and Loamy Clay). Set up the experimental apparatus, collected/analyzed sensor response data, and summarized findings.
- Demonstrated the capability of the sensor to measure soil ammonium in a temporally dynamic manner, with a coefficient of variance of 11% that showed robust stability for in-situ monitoring.

Project: "Dimensional Profiling of Sweat Dispersion Across a Wearable Biosensor." **Summer Intern, Clark Summer Research Program, Collegium V Honors Program, 5/2022-8/2022.**

- Collected and interpreted data on the experiment, testing and recording results. Analyzed data constructing the device to hold the sensor designing the patch worn by the patient. Labeled and organized different samples and medical literature. Presented research at the Summer Platform for Undergraduate Research.
- Competed against 150+ undergraduate and graduate student participants in designing and presenting a research poster explaining experimentation, findings, and applications of the project.
- Earned an award for demonstrated expertise in the topic, as well as strong oration/presentation skills during the SPUR poster symposium upon fulfillment of program requirements.